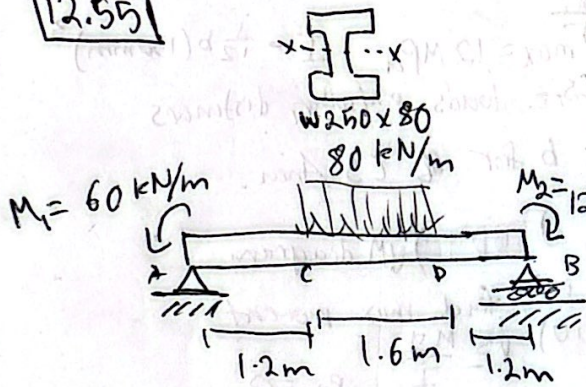


12.55



Given:

W250x80 $\rightarrow I = 126 \times 10^6 \text{ mm}^4$

Forces, moments, pin, roller

find:

VM diagram

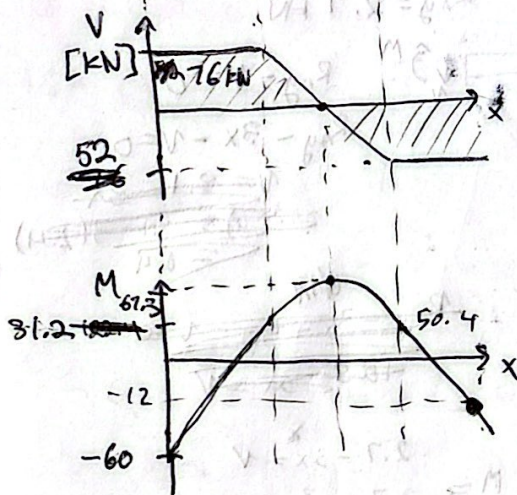
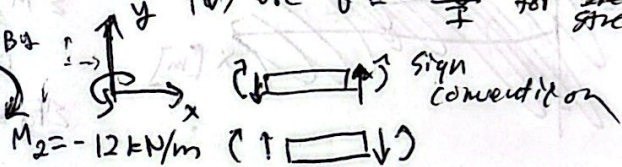
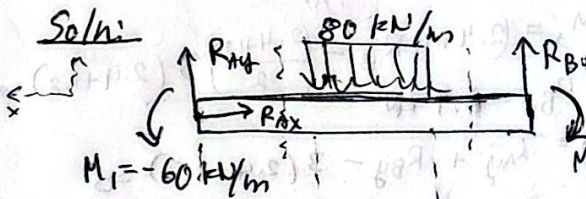
and Max stress

plan:

i) FBD ii) And V iii) find M.

iv) use $\sigma = -\frac{My}{I}$ for stress.

Soln:



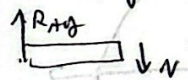
$$\sum F_x = 0 = R_{Ax}$$

$$\sum M_A = (1\text{m})R_{By} - 12 + 60 - (2\text{m})(80 \times 1.6) = 0$$

$$R_{By} = 52 \text{ kN}$$

$$\sum F_y = R_{Ay} + R_{By} - 80(1.6) = 0$$

$$R_{Ay} = 76 \text{ kN}$$



$$\frac{dM}{dx} = V(x) \quad M = \int V(x) dx$$

$$M = 52x + C$$

$$M(0) = +60 \quad 52x +$$

$$\text{max when } \frac{dM}{dx} = 0 \quad V = 0$$

$$V(x) = 76 - 80x \quad x = 0.95$$

$$1.2 \leq x \leq 2.8 \quad x = 0.65$$

$$1.85 \leq x \leq 1.69$$

Max stress.

from graph, max M is 67.3

$$\sigma = -\frac{My}{I}$$

so

$$\sigma = \frac{(67.3 \text{ kN}\cdot\text{m}) \left(\frac{257 \text{ mm}}{2} \right)}{126 \times 10^6 \text{ mm}^4} = -68.63$$

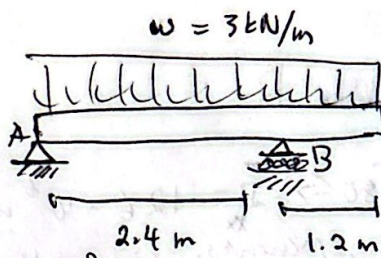
$$= -68.63$$

$$(\sigma_{\max}) = 68.63 \text{ MPa}$$

Reflection:

Looking at the signs of the graph this makes sense. I expect the external moments to counteract the bending from external force. Numbers are within reasonable order of magnitude.

12.61

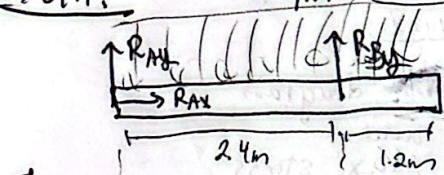


Given: $\sigma_{\max} = 12 \text{ MPa}$ $I = \frac{1}{12} b (150 \text{ mm})^3$

CSA. loads. pin/roller, distances

find b for CS Area.

Soln:



Plan: i) FBD ii) VM diagram

iii) find max moment

$$v = \frac{M_y}{I}$$

$$R_{Ax} = 0$$

$$\sum M_A = (2.4 \text{ m}) R_B - \left(\frac{2.4 + 1.2}{2} \right) 3 (2.4 + 1.2) = 0$$

$$R_{By} = 8.1 \text{ kN}$$

$$\sum F_y = R_{Ay} + R_{By} - 3(2.4 + 1.2) = 0$$

$$R_{Ay} = 2.7 \text{ kN}$$

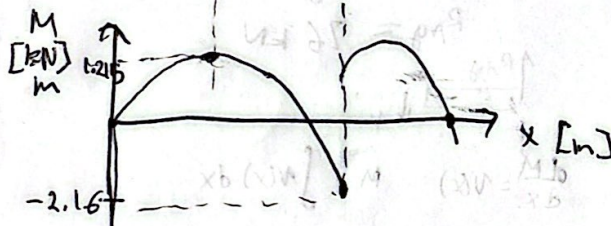
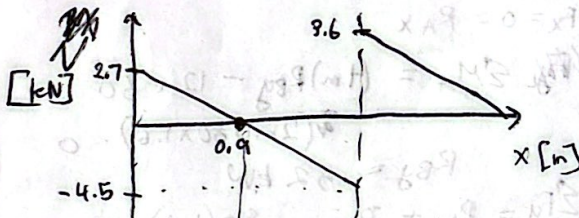


$$R_{Ay} - V = 0$$

$$R_{Ay} - 3x - V = 0$$

$$V = 2.7 - 3x$$

$$V = 0 \Rightarrow 2.7 - 3x = 0 \Rightarrow x = 0.9$$



$$|M_{\max}| = 2.16$$

$$\sigma = \frac{M_y}{I} \quad 12 \text{ MPa} = \frac{(2.16 \text{ kNm}) \left(\frac{150 \text{ mm}}{2} \right)}{\frac{1}{12} b (150 \text{ mm})^3}$$

$$b = 48 \text{ mm}$$

$$M = 2.7x - \frac{3}{2}x^2 + C$$

$$M(0) = 0$$

$$3.6 - 3x = V$$

$$3.6x - \frac{3}{2}x^2 + C$$

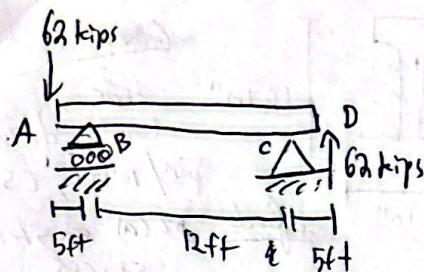
$$M(2.4) =$$

$$- \left(\frac{2.4}{2} \right) 3 (2.4) + (2.4) (8.1) + M = 0$$

Reflection

This order of magnitude for b makes reasonable sense given the height of bar. It is able to pass under elbow - Bernoulli Boundary theorem that states the height should be decreasing larger than width where $150 \text{ mm} > 48 \text{ mm}$, so reasonable here.

12.64



Given: $\sigma_{max} = 24 \text{ ksi}$

Dimensions, pin/roller.

$F = 62 \text{ kips}$

Find:

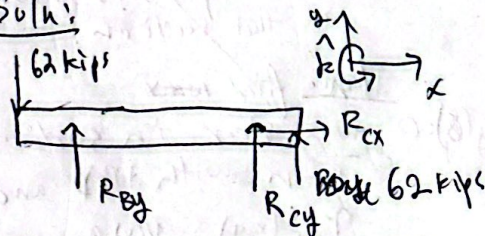
best wide flange beam economically.

Plan:

Draw FBD, find VM diagram to get max Moment. Use $\sigma = -\frac{My}{I}$ and

$\sigma = -\frac{M}{S}$ where $S = \frac{I}{y_{max}}$

Soln:



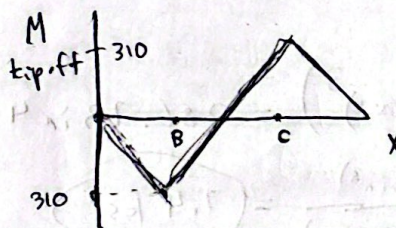
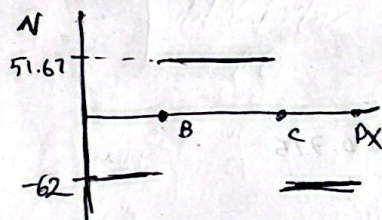
$\sum F_x = R_{cx} = 0$. $\sum M_B = (12+5)F + 12 R_{cy} + 5F = 0$

$\sum F_y = -F + F + R_{cy} + R_{by} = 0$

$R_{by} = 113.67 \text{ kips}$

$R_{cy} = -1.83 F = -113.67 \text{ kips}$

+ convention



~~Shear Force Diagram~~

$\sum F_y = -62 - V(x) = 0$

$V(x) = -62$

$113.67 - 62 - V(x) = 0$

$V(x) = 51.67$

$-62 + 113.67 - 113.67 - V(x) = 0$

$M(0) = 0$

Max moment is $|M| = 310 \text{ kip-ft}$

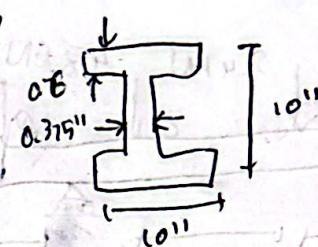
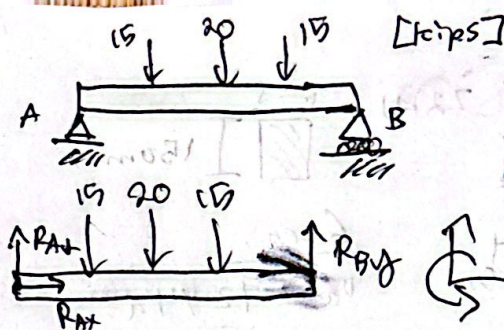
$\sigma = \frac{M}{S}$ $S = \frac{M}{\sigma} = \frac{310 \text{ kip-ft}}{24 \text{ ksi}} = 129.17 \text{ in}^3$

The closest one to 129 is **W27x84** with a $S = 213 \text{ in}^3$

We can't use W24x88 since $154 < 129$.

Discussion: signs make sense w/ diagram and max moment we where I'd expect them to be. 24x84 is reasonable considering the high loads undergoing

13.9



Given:
loads,
dimensions,
pin roller.

Soln.

$$\sum M_B = 15(2) + 20(4) + 15(6) - R_{Ay}(8) = 0$$

$$R_{Ay} = 25 \text{ kip}$$

$$\sum F_y = R_{Ay} + R_{By} - 15 - 15 - 20 = 0$$

$$R_{By} = 25 \text{ kip}$$

expect V diagram to be

so max V is

$$|V_{\max}| = 25 \text{ kip}$$

max moment is

$$|M_{\max}| = 70 \text{ kip ft}$$

max stress

$$\sigma = -\frac{My}{I}$$

$$I = \frac{1}{12}(10)(10)^3 - 2 \left(\frac{1}{12}(5 - \frac{0.375}{2})(10 - 0.6 \times 2)^3 \right)$$

$$I = 286.73 \text{ in}^4$$

$$|\sigma_{\max}| = 41.85 \text{ ksi} \quad 14.68 \text{ ksi}$$

max shear:

$$\tau = \frac{VQ}{It}$$

$$V = 25 \text{ kip} \quad Q = 31.83 \text{ in}^3$$

$$I = 286.73 \text{ in}^4 \quad t = 0.375 \text{ in}$$

$$(\tau_{\max}) = 7.4 \text{ ksi (double)}$$

Compare $\sigma_{\max}$ to $\tau_{\max}$. $\sigma_{\max}$ is much larger than $\tau_{\max}$, so more concerns about stress at ends.

$$\frac{V}{A_{web}} = \frac{25 \text{ kips}}{6.8 \times 0.375} = 6.67 \text{ ksi} \approx \tau_{\max} \quad 6.67 \times 6.70 \text{ pretty good approximation.}$$

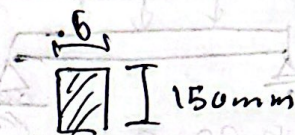
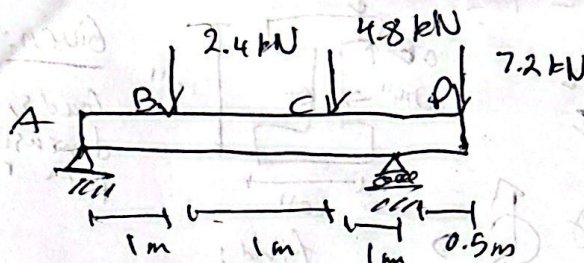
$$7.57 \text{ ksi}$$

Discussion:

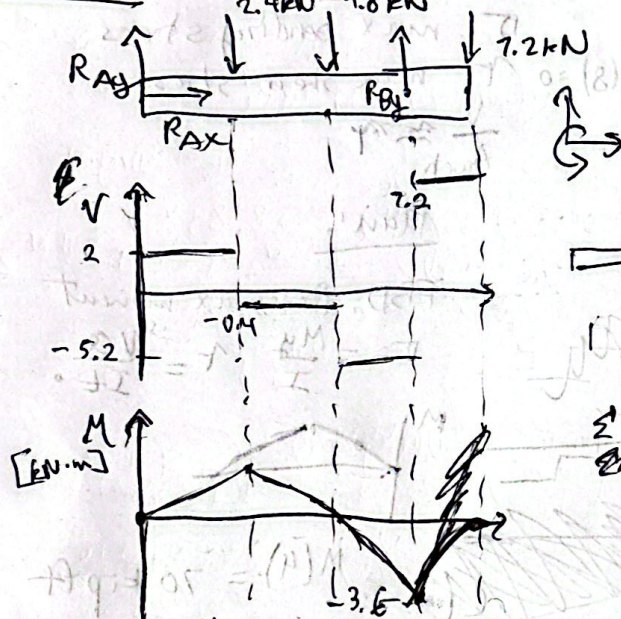
$$\text{error} = 0.1674$$

6.7 and 6.67 being very close is a good indicator of accuracy. Also the locations of max shear is expected.

13.14



Soln:



$$M_{\max} = -3.6 \text{ kN}\cdot\text{m}$$

$$\sigma_{\max} = \frac{-(9.6 \text{ kN}\cdot\text{m})(75 \text{ mm})}{\frac{1}{12}bh^3}$$

$$b = \frac{(3.6 \text{ kN}\cdot\text{m})(75 \text{ mm})}{12 \text{ MPa} \frac{1}{12} (150 \text{ mm})^3}$$

$$b = 80 \text{ mm}$$

$$b = 87.3 \text{ mm}$$

$$b_{\min} = 87.3 \text{ mm}$$

Given:

$$\sigma_{\text{all}} = 12 \text{ MPa}$$

$$\tau_{\text{all}} = 825 \text{ kPa}$$

find minimum b.

Plan:

find max V and M for

$$\sigma_{\text{all}} = \frac{My}{I} \quad \text{and} \quad \tau_{\text{all}} = \frac{VQ}{It}$$

$$\text{Solve for } b \text{ using } I = \frac{1}{12}bh^3$$

$$\sum F_x = R_{Ax} = 0$$

$$\sum M_A = -(1)(2.4) - (2)(4.8) + (3.5)R_{By} - (3.5)(7.2 \text{ kN}) = 0$$

$$R_{By} = 12.4 \text{ kN}$$

$$\sum F_y = R_{Ay} + 12.4 - 2.4 - 4.8 - 7.2 = 0$$

$$R_{Ay} = 2 \text{ kN}$$

$$2 - 2.4 - 4.8 + 12.4$$

$$\sum M_A =$$

$$Q = \int y dA = 0 \quad \bar{Q} = 0.0028125$$

$$V_{\max} = 7.2 \text{ kN}$$

$$b = \frac{(7.2)(0.0028125)b}{\frac{1}{12}(0.15)^3 b}$$

Reflection

these numbers are close which indicate a degree of accuracy. The signs on the graph matters since when imagining how it comes down or bends.

Additional Problem

chuck pin pt 2.

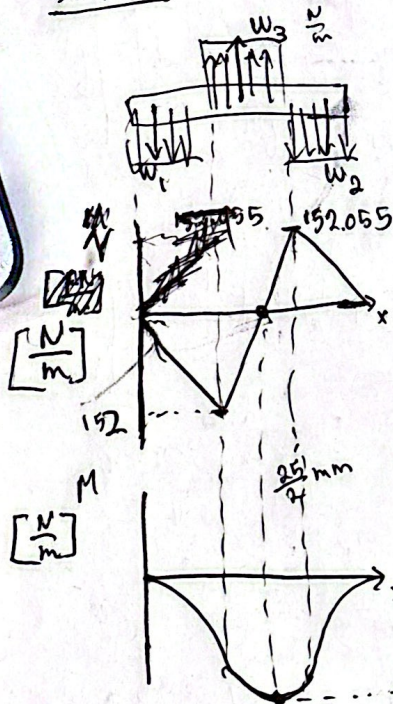
Given
 $w_c = 18 \text{ kg}$
 $w_{ht} = 13 \text{ kg}$
 $E = 200 \text{ GPa}$
 $\sigma_y = 200 \text{ MPa}$
 $\tau_y = 150 \text{ MPa}$

Find
 find min pin diameter
 for $n = 3$ for bending stress,

Plan:

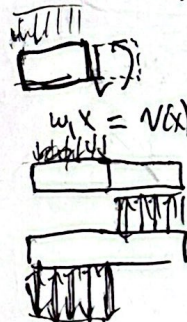
FBD of pin
 $M(x)$ diagram to find max
 $\sigma = -\frac{My}{I}$ and $\tau = \frac{VQ}{It}$,
 find minimum.

Solu:



$$w_1 = w_2 = \frac{(18 + 13) \text{ kg}}{5 \text{ mm} + 5 \text{ mm}} = 30411 \frac{\text{N}}{\text{m}} = 30.411 \frac{\text{kN}}{\text{m}}$$

$$w_3 = \frac{(30411)2(10 \text{ mm})}{10 \text{ mm}} = 20274 \frac{\text{N}}{\text{m}} = 20.274 \frac{\text{kN}}{\text{m}}$$



$$M(5 \text{ mm}) = 152.055$$

$$-w_1(5 \text{ mm}) + w_3 x = 20274$$

$$-152.055 + w_3 x = V$$

$$w_1 x = V(x) \quad \text{then } M_{\max} = 152.055$$

$$\frac{1}{2} w_1 x^2$$

$$M_{\max} =$$

$$-0.95 \text{ N/m} + 152.055 x + \frac{1}{2} w_3 (x)^2 = M_{\max}$$

$$-152.055 \left(\frac{5 \text{ mm}}{2} \right) + \frac{1}{2} w_3 \left(\frac{5 \text{ mm}}{2} \right)^2 = M_{\max}$$

$$M_{\max} = -0.95 \text{ N/m}$$

$$r = \left(\frac{M}{\sigma \frac{1}{4} \pi} \right)^{\frac{1}{3}}$$

$$\sigma_y = -\frac{My}{I} \quad I_x = \frac{1}{4} \pi r^4$$

from table.

$$r^4 = \frac{M}{\sigma \frac{1}{4} \pi}$$

for

$$r = \left(\frac{M}{\sigma \frac{1}{4} \pi} \right)^{\frac{1}{3}}$$

$$r = \left(\frac{M}{\sigma \frac{1}{4} \pi} \right)^{\frac{1}{3}}$$

$$2r = d = 4.64 \text{ mm}$$

$$\tau = \frac{VQ}{It}$$

$$150 \text{ MPa} = \tau = \frac{V \frac{4r}{3\pi} \frac{\pi r^2}{2}}{\frac{1}{4} \pi r^4 2r \sqrt{1 - \frac{8}{9\pi^2}}}$$

$$2r = 3.39 \text{ mm}$$

min diameter is $\boxed{4.64 \text{ mm} = d}$ for limiting

Reflection.

The bending stress is much larger than shear stress. The Oot M is 5mm w/ 5mm and 15mm, the units of the crazy τ eq was fine.

$$\bar{y} = \frac{4r}{3\pi}$$

$$t = \sqrt{r^2 - \bar{y}^2 + \left(\frac{t}{2} \right)^2}$$

$$A = \frac{\pi r^2}{2}$$

$$\sqrt{r^2 - \frac{8r^2}{9\pi^2}} = \frac{t}{2}$$

$$t = 2r \sqrt{1 - \frac{8}{9\pi^2}}$$